RESEARCH FIELD BOOK

DEHRADUN DOON VALLEY IN CRISIS

WATER, CONCRETE AND CLIMATE

Research on springs, monsoon risk, green cover and urban governance in the Doon Valley

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An evidence-led review of water security, urban growth, flood exposure, land-use change and practical responses

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Research report • Dehradun, Uttarakhand

Reader’s Note

Three distinctions are maintained throughout. First, a published finding is not the same as an inference drawn from it. Second, a recommendation is not described as an existing legal requirement unless an official source establishes that status. Third, city-wide or district-wide data are not used to make unsupported claims about every spring, colony or construction site. The term “Doon Valley” is used broadly for the valley landscape around Dehradun. Some studies use the municipal area; others use a wider valley boundary. Figures from these studies should not be combined without checking their geographic extent, time period and method. The legal status of building rules can also change; users should check the current notified text before acting on a specific property or project.

How to Use This Book

- Urban planners can use the chapter structure as a checklist for catchment plans, master-plan reviews and building approval procedures.
- Students can use the references and proposed field methods to design a small research project without confusing observation with proof.
- Local administration can adapt the water-source register, compliance checklist and annual scorecard, provided the responsible departments validate the methods first.

The Central Proposition

Dehradun does not have one water problem. It has linked problems of source reliability, land conversion, drainage, flood exposure, heat, enforcement and public accountability. Solutions will work only when the physical system and the institutions responsible for it are understood together. This book brings together peer-reviewed studies, public planning records, government guidance and reporting This book brings together peer-reviewed studies, public planning records, government guidance and reporting on recent events concerning Dehradun?s urban future. It does not replace site surveys, legal advice, engineering calculations or current official records.

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Introduction. The Valley Is a Hydrological System

A city can grow quickly and still misunderstand the ground it is built on. In Dehradun, the pressure for roads, housing, offices, shops and visitor infrastructure is visible across the valley and the foothill belt. The equally important changes are less easy to see: runoff reaching a drain faster than before, a spring with lower seasonal flow, open land disappearing from a catchment, a roadside channel covered during construction, or a rainwater system installed but never maintained. The crisis is not simply that there is less water or more concrete. It is that decisions made in separate offices can produce a common physical result. A building approval considers one plot. A road project considers a route. A utility considers a supply network. A disaster office prepares for emergencies. Yet the water crosses each boundary. If the systems are not coordinated, responsibility is divided while risk is shared. The land-cover record supports concern about the scale of change. Research using satellite data found substantial urban expansion in Dehradun between 1991 and 2018, alongside declines in dense vegetation and dry river-bed classes; another study of the wider Doon Valley found a marked increase in urban area between 2000 and 2019. [1, 2] These studies are not substitutes for current parcel-level mapping. They are a reason to keep that mapping up to date. The floods of September 2025 made the relationship between rainfall, terrain and exposure immediate for residents. [9, 10] But the event should not be used as a shortcut to a single explanation. Intense rainfall is a hazard; where it causes catastrophic damage also depends on landform, drainage, buildings, river space, maintenance and emergency response. Climate change may alter the probability or intensity of some hazards, but the contribution to a single local event needs specific analysis. This book is organised around practical questions. Where does water come from? What does a spring depend on? How do paved surfaces change runoff? Which buildings and sites need closer scrutiny? What should be measured before a project is called successful? And how can residents provide useful observations without turning an unverified allegation into an official fact? The answer begins with a change in administrative habit: treat water pathways and open land as infrastructure. They should appear in plans, approvals, budgets, inspections and public reporting, not only in environmental chapters that are consulted after a project has been designed.

Chapter 1. The Valley Is a Water System

The first planning error is to treat water, land and buildings as separate subjects.

1.1 A valley shaped by water

Dehradun is often discussed as a city expanding towards the foothills. That description is correct, but incomplete. The wider Doon Valley is a connected physical system: rain falls on ridges and built surfaces, infiltrates soil where conditions permit, moves through weathered rock and valley-fill deposits, emerges at springs, and travels through seasonal channels and rivers. Roads, drains, plots and retaining walls alter parts of this pathway. They do not remove the underlying need for it. The planning boundary of a municipal ward rarely matches the boundary of a spring catchment, a groundwater unit or a drainage basin. Water can enter the ground beyond the place where a well is used. A blocked culvert upstream can flood a road several kilometres away. A new roof can change runoff at the plot edge, while the cumulative effect of hundreds of roofs becomes a ward-level problem. The scale mismatch is not a minor technicality. It affects who should act, which data matter and where public money is spent.

1.2 What research tells us about urban expansion

Remote-sensing studies provide a useful baseline, provided their time periods and study areas are not mixed together. A study of Dehradun municipal area using Landsat imagery for 1991, 1998, 2008 and 2018 estimated that the urban region expanded by about 80.6 per cent over 27 years; it also reported declines in dense vegetation and dry river-bed classes. A separate Doon Valley study covering 2000 to 2019 reported an increase in urban area of 184 per cent. These are different geographies and methods, so their percentages should not be read as if they were one continuous measurement. They point in the same broad direction: built-up land expanded markedly, while several open-land and vegetation categories came under pressure. [1, 2] These findings do not tell us which specific building caused a flood, or whether every lost patch of vegetation had equal hydrological value. Remote sensing identifies changes in surface cover; field surveys and hydrological measurements are needed to explain their effects. A useful public discussion should keep that distinction clear. The evidence is strong enough to justify better land-use monitoring, but it is not a licence to make unsupported claims about every locality.

1.3 The groundwater question is more local than a district average

Doon Valley aquifers include permeable valley-fill deposits commonly described as Doon gravels. Research on groundwater chemistry has shown that these deposits are important aquifers and that groundwater quality reflects both natural interactions with local geology and, in places, human influence. [3] This matters for planning. Rain that reaches an appropriate recharge area can support groundwater storage; contaminated runoff directed into the same area can create a different problem. A district-wide groundwater assessment can be reassuring without describing

the experience of a particular colony, spring or borewell. For example, the 2023 dynamic groundwater resource assessment lists Dehradun's stage of groundwater extraction at about 32.15 per cent. [4] That number should not be misrepresented as proof that every aquifer is secure or that groundwater levels never

decline. The assessment is a broad resource accounting measure. Local pumping concentration, seasonal water tables, water quality and changes in land cover require separate monitoring. A public dashboard should display the scale and limits of every number.

1.4 Define the problem before choosing a solution

The phrase "water crisis" can conceal several different problems: a spring that has stopped flowing, a borewell whose water level has fallen, a household receiving supply only intermittently, a contaminated source, or a street that floods during intense rain. These problems overlap, but they are not interchangeable. A recharge trench may help one catchment and do little for another. A new drain may move water faster without reducing downstream risk. A water tanker may respond to immediate scarcity while leaving its cause untouched. The starting point should therefore be a local water account. For each micro-catchment, record sources, users, seasonal availability, flow or level where measurable, water quality, drainage routes, land-cover change and the agencies responsible. It will be imperfect at first. That is not a reason to delay it. A transparent baseline that is improved every year is more useful than a polished map that nobody updates.

Chapter 2. A Spring Is More Than Its Outlet

To restore a spring, first understand the area that feeds it.

2.1 The spring and its recharge area

A spring is the visible point at which groundwater reaches the surface. Its discharge depends on geology, slope, fractures or permeable deposits, rainfall, vegetation, land use and the condition of its recharge area. Protecting the outlet while ignoring the land that supplies it is unlikely to be enough. Similarly, planting trees near a spring does not guarantee that its flow will recover. The work must begin with a reasoned understanding of where the water comes from and how it moves. Mussoorie offers a nearby research example. A 2021 study selected six gravity springs and their springsheds in the Mussoorie hills and combined geospatial tools with geophysical methods to characterise recharge conditions. [5] The central lesson is methodological: map the springs, identify likely recharge zones, investigate subsurface conditions, and then choose interventions. Mussoorie is not a recipe to be copied without adjustment. The underlying geology and settlement pattern of each site must guide the design.

2.2 Learn from springshed practice, not just construction works

The NITI Aayog resource book on springshed management for the Indian Himalayan Region brings together practices on spring inventories, long-term monitoring, community participation, hydrogeological assessment, implementation and impact measurement. It also recommends consistent formats and publicly accessible repositories so that individual projects contribute to a wider evidence base. [6] The Department of Water Resources' spring-census guidance makes a related point: there is a need for uniform methods to record and map springs and their recharge areas. [7] For Dehradun, a spring-revival project should therefore be designed in stages. First locate and verify the source. Then monitor its discharge through dry and wet periods. Delineate the recharge area with local hydrogeological expertise. Identify land uses that may reduce infiltration or threaten water quality. Only after this assessment should teams decide whether to use contour trenches, small recharge structures, vegetation restoration, fencing, drainage improvements or another measure. Some sites will need several small interventions; others may need protection from contamination before recharge work is appropriate.

2.3 Monitoring should outlast the inauguration

A ribbon-cutting cannot establish that a spring has recovered. The baseline must include at least seasonal discharge, water quality, user demand and the condition of recharge measures. Where feasible, observations should be repeated monthly and after major rainfall events. The record should note whether measurements were taken at the same point, with the same method, and by whom. Otherwise apparent improvements may reflect differences in measurement rather than a hydrological change. Communities can participate directly, but local observers need training and

a clear route for reporting concerns. A spring committee could maintain a simple logbook, while a technical partner periodically checks the method. If discharge improves, the evidence should show by how much, over what period

and compared with what baseline. If it does not improve, the project should document why. Failure is information, not something to remove from the report.

2.4 Do not transfer a hill-spring model blindly to the urban valley

The Rajpur and Sahastradhara belt contains foothill slopes, streams, roads, settlements and changing land uses. Parts of the wider Doon Valley depend on aquifers in valley-fill deposits; not every local water source is a conventional mountain spring. The technical vocabulary should follow the hydrogeology of the site. A spring recharge plan, an urban stormwater plan and a groundwater abstraction plan may need to be coordinated, but they are not the same intervention. Recent state-level work also suggests that the topic is receiving renewed institutional attention. In February 2026, Swami Rama Himalayan University reported a workshop in Mussoorie organised with IUCN and the state Spring and River Rejuvenation Authority, covering source conservation, community-based management, recharge interventions and water quality. [8] This is a reason to engage with existing expertise and programmes, not to assume that every proposed intervention has already been evaluated or scaled successfully.

Chapter 3. When Monsoon Rain Becomes a Disaster

Rainfall is a hazard. The scale of damage also depends on exposure, drainage and land decisions.

3.1 The September 2025 event and what it does - and does not - prove

In September 2025, intense rainfall triggered destructive flooding and landslides in Dehradun and nearby areas. News reports identified Sahastradhara, Maldevta, Manjayada and Karligad among the severely affected locations, with damage to homes, roads, bridges and other infrastructure. [9] Early accounts used the word “cloudburst”; a separate report noted that the India Meteorological Department said the rainfall rate cited for Dehradun did not meet the formal cloudburst threshold. [10] Terminology matters because an account of a disaster should distinguish eyewitness description, operational reporting and a meteorological classification. One event cannot, by itself, quantify the contribution of climate change, illegal construction or drainage failure. It can reveal where exposure is high and where emergency systems were tested. Attribution of a specific event requires a different body of evidence. The practical planning question remains urgent even when attribution is uncertain: why did water and debris reach these places, which routes did they take, and which decisions increased or reduced the consequences?

3.2 Runoff changes when the ground is sealed

Vegetated soil, open ground, wetlands and permeable surfaces can store or infiltrate part of the rainfall they receive, subject to soil condition, slope, antecedent moisture and rainfall intensity. A paved yard or a building roof sends a larger share of water into drains or street channels more quickly. The effect of one property may be small. The combined effect across a growing catchment can be substantial, particularly when drainage is undersized, blocked or routed through a constrained channel. It is tempting to respond to every flood by building a larger drain. But where should the drain discharge? If it delivers water faster to a downstream bottleneck, one locality may be protected at another’s expense. Urban flood planning needs a catchment model that includes natural channels, road culverts, storm drains, cross-drainage structures, depressions and the places where water already accumulates. Maintenance is as important as engineering capacity.

3.3 Keep river space and drainage paths functional

Seasonal streams and urban rivers require space to carry water and sediment, including during unusual events. Encroachment can reduce conveyance, expose buildings to damage and make restoration more difficult. Yet an accurate risk map must distinguish the active channel, floodplain, erosion-prone bank, drainage reserve and land that may be developed under specific conditions. These are technical and legal determinations; a satellite image alone is not a final boundary survey. The Rispana and Bindal rivers were part of public debate following the

September 2025 disaster, including disagreement about a proposed elevated corridor project. [11] That debate illustrates the need to assess transport benefits alongside hydrology, flood safety, displacement, ecology and long-

term maintenance. An infrastructure proposal should publish the relevant technical studies and alternatives, not rely only on travel-time benefits or broad environmental slogans.

3.4 A layered flood programme

A realistic urban flood programme has several layers: reliable rainfall and water-level observations; routine desilting and waste removal; inspection of culverts and bridges; protection of natural drainage lines; restoration of upstream infiltration where scientifically suitable; permeable public surfaces; safe overflow routes; and warnings that reach people before roads become impassable. Emergency shelters and evacuation plans must include older people, children, people with disabilities and households with limited transport. No single intervention makes a catchment flood-proof. The standard should be risk reduction and lower losses, measured over time. Maps should show not only assets built but also whether blocked drainage points declined, warning time improved, critical services stayed open and repeat damage fell. That is a harder account to produce. It is also the one that helps planners learn.

Chapter 4. Concrete Along the Rajpur - Sahastradhara Belt

The question is not whether growth will occur, but where it is safe and under what conditions.

4.1 Read the belt as a connected development area

The Rajpur - Sahastradhara corridor is not a uniform strip of land. It includes road-front commercial activity, residential colonies, foothill terrain, local drains and streams, institutional properties, open areas and ecologically sensitive patches. Slope, geology, access roads and drainage vary from site to site. A generic rule that treats the entire belt as either developable or undevelopable will miss those differences. Planning needs parcel-level review nested inside a catchment-level assessment. The pressure to build is shaped by housing demand, commercial opportunity, road access, tourism and expectations about future infrastructure. These are real economic forces. But private gains from a new building may not include the cost of added runoff, road congestion, water demand, downstream sediment or public drainage works. Planning regulation is how some of those wider costs are considered before construction, rather than after damage has occurred.

4.2 Illegal plotting is a governance signal, not a complete map

The Mussoorie Dehradun Development Authority publishes sealing and demolition actions, including orders relating to illegal plotting near Sahastradhara Road and unauthorised construction in the IT Park area. [12] Those entries establish that enforcement actions have occurred at named sites. They do not tell us the total prevalence of unauthorised development across the corridor, nor do they settle every question about a particular property. A defensible assessment should combine approved-layout records, satellite imagery, site inspections, complaint history and current legal status. Public maps should use carefully defined categories: approved, under review, notice issued, order passed, appeal pending, demolition or remediation completed. An allegation should not be displayed as a final finding. The same database should record the date of the last inspection and the authority responsible for the next step. Otherwise public attention may rise during a controversy and vanish before the site’s final status is known.

4.3 Slope, water and cumulative impact

A single building plan can satisfy a narrow plot-level test and still contribute to a wider cumulative problem if many similar projects are approved without considering their combined effects. Roads alter runoff and may cut across natural drainage. Retaining walls can be necessary, but poorly designed or poorly drained retaining structures can create additional risk. Excavation, hill cutting and spoil disposal should be assessed in relation to slope stability, runoff paths and the load placed on local infrastructure. A planning application in the foothill belt should therefore answer more than “Is the land privately owned?” It should show the approved land use, slope and geotechnical assessment where needed, the

receiving drainage route, access for emergency vehicles, water supply plan, wastewater connection, tree and stream constraints, construction debris arrangements, and the consequences of a blocked outlet. If the applicant cannot demonstrate how water leaves the site safely, the plan is incomplete.

4.4 Make compliance visible before and after approval

A sustainable development system requires reliable records at four points: application, permission, construction and occupancy. The approved plan must be the plan checked on site. Drainage, rainwater harvesting, pervious area and slope-protection requirements should be inspected before the occupancy certificate, not merely listed on drawings. Deviations should trigger a documented response with a time limit and an appeal route. The purpose of enforcement is not to produce a few dramatic demolition photographs. It is to make the rules predictable. When identical violations receive different outcomes, compliant builders are disadvantaged and public confidence falls. A public quarterly report could disclose inspections, violations, notices, resolved cases, unresolved cases and the time taken at each stage. Sensitive personal information need not be published; the status of public enforcement should be.

Chapter 5. Green Cover Is Infrastructure

Trees and open land matter, but the category “green” is too broad to guide every decision.

5.1 What the satellite record can show

The Dehradun land-cover studies do more than document a visual change. They show why urban growth should be monitored with repeatable spatial data. The 1991 - 2018 municipal-area analysis reported an 80.6 per cent increase in urban area and a sharp decline in dense vegetation and dry river- bed classes. A separate analysis of the wider Doon Valley found significant expansion of urban area between 2000 and 2019. [1, 2] The precise estimates depend on the study boundary, classification method and image dates, so planning documents should cite those details whenever they reproduce a figure. Satellite classification cannot tell us everything about a tree. It may not distinguish native forest from a landscaped lawn in a way that captures ecological value; seasonal images can also change the apparent condition of a site. A land-cover map needs ground verification, canopy assessment, species information and records of whether a green area is publicly accessible, privately held or legally protected.

5.2 Urban heat is a measurable issue

A peer-reviewed study using Landsat data for Dehradun in 1991, 1998, 2008 and 2018 found substantial changes in land cover and surface temperature. It estimated an 80.6 per cent expansion of the urban region over 27 years and reported that mean land-surface temperature had increased by more than 9 degrees Celsius across seasons in its analysis. [13] This is a satellite-derived surface-temperature result, not a claim that every resident’s air temperature rose by the same amount. The distinction should remain explicit. For urban design, the implication is straightforward: shade, canopy, reflective surfaces, ventilation, water-sensitive open space and building orientation influence exposure to heat. The value of a tree is not reducible to the number planted. Survival after three or five years, crown size, species suitability, rooting space and maintenance determine what the tree eventually provides.

5.3 Protect connected open space, not only decorative patches

Small parks are useful, but open land also performs less visible functions: it can preserve a drainage route, support infiltration, provide soil stability, retain biodiversity or keep a slope from being intensively disturbed. A development review should identify these functions before categorising a plot as “vacant”. Vacant land in a cadastral record may still be important to public safety or water security. Green-space policy should include a hierarchy. First avoid damage to valuable and difficult-to-replace areas. Then minimise unavoidable disturbance. Restore the specific functions that were affected. Only after those steps should any offset be considered, and an offset should not be used to justify loss of a site whose ecological or hydrological value cannot realistically be reproduced elsewhere.

5.4 Measure canopy and survival, not announcements

A ward tree register should record location, species, date planted, approximate canopy, survival, watering responsibility, condition and risk to nearby infrastructure. It should link to a maintenance budget. Public reporting could show canopy change across several years, with separate figures for planted saplings and surviving trees. A planting drive is an activity; a maintained canopy is an outcome. Residents can help record mature trees and flag cutting, but the authority should retain responsibility for verification and enforcement. Crowd-sourced reports can be mistaken or duplicated. Their value improves when every report has a location, date, supporting photograph where safe, a review status and a mechanism to correct errors.

Chapter 6. Building By-Laws Must Work on the Ground

A rule on paper changes the city only when it is clear, inspected and enforced.

6.1 Start with the current legal text

MDDA and the Uttarakhand Housing and Urban Development Authority publish building by-laws, amendments and map-approval material on their official websites. [14, 15] The applicable provisions depend on the current consolidated text, the development plan, the plot, building use and any relevant environmental or judicial restrictions. A draft document or an older checklist should not be treated as the law in force. Before acting on a specific property, planners and owners should verify the latest notified provisions with the responsible authority. This book does not give legal advice and does not claim that every proposed safeguard below is already a binding requirement. Instead, it identifies provisions that an updated, risk-sensitive system should make explicit and auditable.

6.2 Proposed water-sensitive checks

A building approval should require a site-specific water and drainage statement. It would identify roof runoff, surface runoff, the expected discharge point, existing natural drainage, on-site storage or recharge provisions, wastewater treatment and the risk of contaminating groundwater. The authority should specify which calculations, drawings and certificates are required for each building category. Requirements must be scaled to site and building characteristics rather than copied mechanically from an unrelated city. Rainwater harvesting should not be reduced to a tank drawn on a plan. The system needs a collection surface, conveyance, first-flush arrangements where appropriate, filtration, overflow routing, safe storage or a scientifically suitable recharge mechanism, and access for maintenance. Recharge is not safe in every location. Where groundwater contamination or unstable ground is a concern, collected water may need to be stored and reused rather than injected or directed underground.

6.3 Proposed slope, tree and runoff safeguards

For foothill and sloping sites, approval should trigger a proportionate geotechnical review where slope, excavation or local conditions warrant it. The plan should show cut-and-fill balance, retaining structures, subsurface drainage, construction spoil management and emergency access. A stream or natural drainage alignment should be surveyed before it is diverted, covered or crossed. The review should consider cumulative changes in the receiving catchment, not only the individual plot boundary. The development plan should also record existing trees and vegetation, proposed removals, replacement commitments and responsibility for aftercare. Permeable open space needs a clear definition and on-site verification. A thin strip of decorative gravel over an impermeable base should not be counted as meaningful infiltration area unless it performs that function under the approved design.

6.4 Inspection, occupancy and public accountability

Plans become effective only when construction is checked. A risk-based inspection system could prioritise work near streams, steep slopes, sensitive recharge areas and known flood corridors. It should use a dated inspection record, geotagged photographs, a standard checklist and a clear escalation path. The same system should inspect compliant projects and alleged violations so that enforcement does not depend entirely on who files a complaint. At completion, critical systems should be tested: the rainwater harvesting installation, the route for overflow, the pervious area, drainage connections and any required slope-protection measures. The occupancy certificate should reflect what was actually built. After a major change to the property or its use, relevant parts of the approval should be reviewed again. Rules are most credible when good-faith applicants can understand them and when non-compliance has predictable consequences.

Chapter 7. Bring Rainwater Harvesting Back into the Mainstream

Capture, store, reuse and recharge only after checking what is safe and useful at each site.

7.1 Calculate the roof before buying the tank

A rainwater system begins with a basic water balance. The approximate volume that can be collected depends on rainfall over the chosen period, roof area, collection efficiency and losses. The design should then compare likely supply with demand: toilet flushing, cleaning, gardening or other permitted uses. A tank that is too small may overflow through much of the monsoon; a tank that is too large may cost more than the realistic demand justifies. Actual rainfall records and the building's use should shape the calculation. The first design question is often overlooked: what happens when the tank is full? Overflow should follow a safe route that does not flood a neighbour, erode a slope or overload a road drain. The system also needs a way to remove leaves and sediment, inspect gutters, clean storage and prevent mosquito breeding. Without maintenance, a rainwater installation can become a liability rather than a reliable supply.

7.2 Storage and recharge are different choices

Storage retains collected water for use. Recharge aims to allow water to enter the subsurface and support groundwater storage. Both may be beneficial, but they have different design requirements and risks. Stored water needs an appropriate cover, access for cleaning and a use plan. Recharge requires a sound understanding of soil, geology, groundwater quality and local flow paths. Water contaminated by roof debris, industrial residues or unsuitable surface runoff should not be directed blindly into recharge structures. The older idea that all rainwater should be pushed underground is too simple. In some parts of the valley, soil infiltration may be restricted; in others, recharge may be possible but requires pretreatment or separation from polluted runoff. A competent assessment should say which method is appropriate, what evidence supports it, how overflow is managed and who will inspect the system.

7.3 Retrofit public buildings first

Schools, colleges, hospitals, government offices, markets and parking facilities can offer useful demonstration sites because their roofs and water bills are often easier to document than those of dispersed private homes. Each site should begin with a survey of roof condition, current consumption, plumbing, space for storage, drainage and likely uses. The project should publish its design, capital cost, annual maintenance cost and performance after the first and second monsoons. A public building can also serve as a teaching site. Students can measure rainfall, tank levels and non- potable water use under supervision. The aim is not to collect attractive photographs but to show

whether the system saves treated supply, reduces runoff or improves a suitable recharge pathway. The measurements determine which designs should be repeated.

7.4 Maintain systems as public infrastructure

A harvesting asset needs a named owner. For a private building that is the owner or building manager; for a school it may be a designated facilities officer; for a shared complex it should be a committee with a maintenance budget. The records should include inspection dates, cleaning, repairs, water use, overflow problems and complaints. A working system that nobody can maintain is not a sustainable system. The municipal authority can help through standard drawings, technical support, vetted contractors, training and periodic audit. Incentives should reward verified performance rather than installation alone. A rebate tied to inspection and continued operation may be more effective than a one-time payment for a structure that is never used.

Chapter 8. Let Citizens Map the Water They Live With

A public inventory is valuable when residents can contribute and authorities can verify.

8.1 Build a common register

A citizen-led water map should include springs, wells, hand pumps, streams, ponds, drains, flooding points, rainwater-harvesting systems and visible changes in water availability. Every entry needs a unique identifier, coordinates or a carefully described location, date observed, source type, seasonality, condition, use and verification status. If a source is no longer functioning, the record should remain with that status instead of being deleted. Historical information is often essential to understanding change. The water-source inventory should be consistent with national guidance. The Department of Water Resources notes that systematic spring and springshed mapping needs standard formats and methods, and the NITI Aayog resource book recommends inventory systems, long-term monitoring and accessible repositories. [6, 7] Local mapping can contribute to this process if it records its methods and uncertainty.

8.2 Use a simple observation protocol

A first visit should record the source location, photographs where safe, current condition, date and time, recent rainfall if known, visible flow or water level, signs of contamination, nearby construction and the person or group reporting the observation. Volunteers should not enter unsafe channels, trespass on private property or handle suspected contaminated water. Water-quality claims require suitable tests; colour, smell and clarity alone do not establish potability. Repeated visits matter more than one dramatic observation. A dry spring photographed in May and again in October tells a different story from a source recorded only once. Volunteers can record qualitative categories consistently, while trained teams take discharge measurements or collect samples using an approved method. The database should state the evidence level for each field.

8.3 Pair local knowledge with technical verification

Residents may know that a tap once flowed year-round, that a seasonal channel changed direction after road work, or that a public tank has been disconnected for years. This knowledge can guide investigation, but it should not be silently converted into a measured scientific result. A good workflow preserves the resident’s account and labels it as such, then assigns an agency or technical partner to verify the claim where a decision depends on it. The system must also allow corrections. A resident should be able to flag a duplicate source, a misplaced point or an outdated status. Corrections should be logged rather than erased. This protects the credibility of volunteers and prevents an early error from becoming permanent official data.

8.4 Protect people while opening the data

Public transparency does not mean publishing every detail. The map should not disclose personal telephone numbers, household water vulnerabilities or sensitive private-property information without

a clear reason and permission. Aggregated data can be public, while the identity of individual reporters remains protected. Complaints about a named property should move through a formal review process rather than being presented as proven fact on a public map. A useful interface has four states: reported, awaiting verification, verified and resolved or closed. It should show who is responsible for follow-up and when the status was last changed. Citizen science works best when the public can see what happened after the report was filed. Otherwise participation becomes a request for unpaid data collection without meaningful influence.

Chapter 9. Climate Risk Without Easy Answers

A warming climate can change the conditions of risk; local evidence is still needed to explain each disaster.

9.1 Separate climate trends from individual-event attribution

The September 2025 floods make clear that Dehradun can experience rainfall intense enough to cause severe damage. They do not, on their own, prove that climate change caused a particular flood or quantify how much it contributed. Attribution requires observations, physical understanding and, where possible, comparison between actual and counterfactual climate conditions. Local land use, topography, antecedent wetness, drainage and exposure remain relevant even when climate hazards change. This caution should not be mistaken for inaction. Urban systems should be robust to plausible increases in heavy rainfall and variability, particularly where failures have high consequences. Risk planning should use official rainfall records, current intensity-duration-frequency information where available, updated flood histories and scenario testing. The uncertainty itself needs to be visible in the design assumptions.

9.2 Compound risks are harder to manage

The city's water risks are linked. A dry pre-monsoon period can increase household demand and make a declining spring more visible. Intense rainfall may then overwhelm drains, damage roads or contaminate a source. A heat wave can increase demand for water and shade. A landslide may interrupt access to a source or treatment facility. These are not identical hazards, but service planning should ask how one disruption affects the others. A climate-resilience review should therefore test more than one scenario. What happens if intense rain coincides with blocked culverts? If a water source is contaminated while the main road is closed? If a heat wave occurs during a supply interruption? Which services need backup power, emergency storage, a second access route or a public communication plan? These questions translate climate concern into operational preparation.

9.3 Build for uncertainty and allow maintenance

Infrastructure designed only for the most convenient forecast is likely to age poorly. At the same time, oversizing every asset is neither affordable nor necessarily safe. A risk-based approach identifies the consequences of failure, evaluates multiple design conditions, provides safe overflow or shutdown arrangements, and creates a plan to inspect and adapt the asset as new information appears. Nature-based measures, such as conserving vegetation and restoring suitable infiltration areas, can support adaptation, but they are not substitutes for safe drainage, building standards or emergency services. The term should describe a measurable function, not serve as a marketing label. A restored channel, a permeable public courtyard and an urban tree canopy have different functions, maintenance needs and limits.

9.4 Communicate uncertainty honestly

Public communication should distinguish observed fact, modelled projection, expert interpretation and policy choice. For example, a measured rise in local surface temperature is an observation from a specified method; the cause of a single flood may be uncertain; a proposed building restriction is a governance decision. Mixing these categories can undermine trust even when the underlying concern is real. An annual climate and water report for Dehradun could include heavy-rainfall days, flood locations, damaged assets, water-source status, seasonal groundwater observations, canopy cover, heat exposure and the progress of adaptation projects. The first edition may expose data gaps. Publishing those gaps is part of building a credible record.

Chapter 10. A Five-Year Programme for a Water- Sensitive Doon

The shift from isolated projects to public accountability is the central test.

10.1 Year One: establish the baseline

The first year should create a shared geographic and institutional baseline. Map springs and other water sources, streams, drains, flood reports, public rainwater systems, green cover, construction permissions and known enforcement cases. Compile available groundwater and rainfall observations and document their limits. The aim is not to delay action until every data field is perfect. Immediate safety and maintenance work should continue while the baseline is built. A multi-agency group should assign responsibility for each layer: development authority, municipal corporation, water utility, disaster management, forest and watershed agencies, district administration, academic partners and resident groups. Every dataset should have a named custodian, an update interval and a process for corrections. Without this, multiple departments may publish overlapping maps that cannot be reconciled.

10.2 Years One and Two: select catchment pilots

Choose pilot catchments that represent different conditions rather than selecting only the easiest locations. One could focus on a spring and its recharge zone; another on a dense urban drainage network; a third on the Rajpur - Sahastradhara development belt. Selection criteria should include exposure, feasibility, community interest, data availability and the potential to learn something that can be applied elsewhere. Each pilot should publish a problem statement, baseline, responsible agency, costs, expected outcomes, maintenance plan and public reporting schedule. Measures should be tested at manageable scale. For a spring, monitor discharge and quality; for a flood-prone catchment, assess waterlogging frequency and drainage blockages; for a construction corridor, track compliance, runoff management and unresolved violations.

10.3 Years Two and Three: make rules and budgets consistent

Once the baseline identifies repeated problems, the authority can revise technical guidance and operating procedures. Building approvals, road projects, drainage maintenance, tree management and spring revival should use compatible maps and assumptions. Capital budgets must include inspection and maintenance rather than treating them as optional future tasks. Where new evidence conflicts with an earlier plan, there should be a documented process to review the decision. Public procurement should include performance checks and safeguards against conflicts of interest. Projects should be evaluated on whether they deliver the intended function, not merely whether the contracted structure was completed. An independent technical review is appropriate for high-risk projects near rivers, steep slopes or critical water sources.

10.4 Years Three to Five: evaluate and scale what works

Scaling should depend on measured results. A recharge measure should be expanded only when the hydrogeology supports it and monitoring shows a credible benefit. A rainwater system should be replicated when water use, maintenance and overflow performance justify the design. A new enforcement protocol should be assessed by the resolution of cases and the consistency of inspection, not by the number of notices alone. A five-year programme should publish negative as well as positive results. If a pilot fails, record the reason: unsuitable geology, poor maintenance, weak participation, poor design, budget interruption or inaccurate assumptions. This is how public systems improve. The alternative is a cycle of announcements, fresh schemes and repeated problems.

Conclusion. A Valley That Can Keep Its Water

Dehradun does not need to choose between a functioning city and a protected valley. It does need to stop treating the costs of urban growth as someone else's responsibility. Water supply, drainage, construction permissions, tree cover, road engineering and disaster management are connected by the places where water moves and the land that stores it. The first priority is not a grand announcement. It is an accurate baseline: which sources still flow, where seasonal water accumulates, which drains fail, where construction has changed runoff, what land cover has been lost, and which rules apply. The second is a clear division of responsibility. Each data layer and each intervention needs an owner, a maintenance budget and a date for review. The third is public accountability. Residents should be able to see whether reports were checked, whether violations were resolved and whether projects delivered the function promised in their approval documents. Mussoorie's springshed research offers a useful reminder that source restoration begins with understanding the recharge area, not with installing a structure. Rainwater harvesting offers a practical route to reduce demand and manage runoff, provided that systems are safe and maintained. The land-cover studies show why green space must be measured as more than decoration. Public building records show that enforcement action exists; the challenge is to make compliance predictable, timely and verifiable. None of these steps removes the uncertainty of future rainfall or guarantees that no flood will occur. That is not the standard by which a serious city should be judged. The standard is whether it reduces avoidable risk, protects the sources on which people depend, learns from failure and makes decisions in the open. A valley under pressure still has options. The question is whether those options will be used before the next emergency forces the issue.

Use one record per source. Keep a stable source ID so the same location can be revisited over multiple seasons. Field What to record Source ID and type Unique ID; spring, well, hand pump, stream, pond, tank, drain or rainwater system. Location Coordinates or precise public description; catchment or ward; avoid publishing sensitive personal information. Observation Date, time, observer role, weather/rainfall if known, photograph where safe. Seasonality Flow/level observed; seasonal, perennial, intermittent or inactive; method used. Use and users Drinking, household use, irrigation, public use or ecological function; estimate only where defensible. Water quality Visual observation separated from lab-tested parameters; record sampling date and laboratory/method. Land condition Nearby construction, vegetation change, drainage, dumping, erosion or possible contamination. Verification Reported, awaiting verification, verified, referred, resolved; responsible office and date of update. Maintenance Who is responsible, last inspection, next inspection, known repairs and unresolved issues.

1. Observe from a safe, publicly accessible place. Do not enter fast-flowing water, unstable ground or private property without permission. 2. Record the date and location before describing the problem. Use neutral language: “water was not flowing during this visit” is stronger than “the spring has permanently dried up” unless the longer- term record supports it. 3. Keep testimony, visual observations, measurements and laboratory results in separate fields. 4. Do not publish personal telephone numbers, household vulnerability details or unverified accusations about named individuals. 5. Submit reports through a channel with a tracking number, verification status and a public response deadline. 6. Return to the same location during a different season and after major rainfall where it is safe to do so.

- Current land-use status and latest applicable notified by-laws checked by the competent authority. Site plan shows slopes, trees, natural drains, nearby water sources, access and receiving stormwater system.
- Runoff statement identifies roof and ground runoff, safe overflow routes and effects on adjoining properties. Rainwater harvesting design includes realistic end use or a hydrogeologically suitable recharge design, plus maintenance access. Excavation, retaining works, cut-and-fill and construction spoil are assessed proportionately to site risk. Pervious area and planting commitments are clearly defined, measurable and inspectable on site. Wastewater and solid-waste arrangements prevent contamination of drains, streams and recharge areas. Inspection record compares the as-built property against the approved plan before occupancy. A named party and recurring budget are responsible for maintenance.

Indicator Suggested annual record Sources Number of mapped sources; share visited in both pre-monsoon and post-monsoon periods. Spring condition Seasonal flow observations, quality results where tested, recharge work and maintenance status. Flood and drainage Reported waterlogging points, repeat locations, blocked culverts, repairs and warning coverage. Rainwater systems Installed, functional, inspected and maintained systems; measured reuse where available. Green cover Tree canopy or land-cover change from a consistent method; survival after 1, 3 and 5 years. Building compliance Inspections, violations, notices, resolved cases, time to resolution and appeal status. Heat exposure Priority locations, shade provision, vulnerable facilities and tree survival. Public response Citizen reports received, verified, resolved, overdue and corrected. Budget and upkeep Capital spending, maintenance allocation, funds spent and assets overdue for repair. These are proposed indicators, not a claim that the relevant data are already collected. Before adopting the scorecard, the responsible agencies should define each indicator, name a custodian, set a consistent baseline and publish the limitations.

Five Living Scholars and Practitioners to Read

The people below are included as starting points for further study because their public work overlaps with water management, springs, ecology or environmental policy. Their approaches differ, and inclusion is not an endorsement of every position they may take. The book uses an original analytical voice rather than imitating any individual author. Sunita Narain. Director General of the Centre for Science and Environment; relevant work includes community water management, rainwater harvesting and environmental policy. Read for the connection between water literacy, equity and public regulation. Official profile / source [16] Dr. Anil Prakash Joshi. Founder associated with HESCO in Uttarakhand; his work links environmental science, local resources, appropriate technology and ecology-based development. Official profile / source [17] Dr. Rajendra Singh. Water conservation practitioner and chairperson of Tarun Bharat Sangh; relevant for community-led water restoration, traditional harvesting structures and public water literacy. Official profile / source [18] Prof. Durgesh Pant. Uttarakhand science and technology leader and co-author of a 2025 review of springs in Uttarakhand; relevant to science communication, technology systems and regional water research. Official profile / source [19] Dr. G. C. S. Negi. Researcher in forest ecology, watershed management and environmental hydrology; co-author of a 2025 review on the state of knowledge about Uttarakhand springs and rejuvenation. Official profile / source [20]

Read across disciplines. A hydrologist can explain recharge and water quality, but a planning researcher may be needed to understand land conversion. A community water practitioner can show how local institutions maintain a source, while an environmental-policy researcher can help assess who bears costs and who benefits. For Dehradun, the strongest analysis will combine these perspectives with current field measurements and official local records.

Selected References and Public Sources

[1] Mishra, K. and Garg, R. D. (2023). “Assessing variations in land cover-land use and surface temperature dynamics for Dehradun, India, using multi-time and multi-sensor Landsat data.” Environmental Monitoring and Assessment, 195. https://doi.org/10.1007/s10661-023-10945-z [2] “Assessment of urban growth effects on green space and surface temperature in Doon Valley, Uttarakhand, India.” Peer-reviewed study of urban expansion from 2000 to 2019 using Landsat data. [3] Dudeja, D., Bartarya, S. K. and Biyani, A. K. (2011). “Hydrochemical and water quality assessment of groundwater in Doon Valley of Outer Himalaya, Uttarakhand, India.” Environmental Monitoring and Assessment, 181, 183-204. https://doi.org/10.1007/s10661-010-1823-7 [4] Central Ground Water Board (2023). Dynamic Ground Water Resources of India, 2023. District-level estimates for Uttarakhand, including Dehradun. [5] Bagchi, D. and colleagues (2021). “A study on spring rejuvenation and springshed characterization in Mussoorie, Garhwal Himalaya using an integrated geospatial-geophysical approach.” Remote Sensing Applications: Society and Environment, 23, 100588. https://doi.org/10.1016/j.rsase.2021.100588 [6] Rathod, R. et al. (2021). Resource Book on Springshed Management in the Indian Himalayan Region: Guidelines for Policy Makers and Development Practitioners. IWMI, NITI Aayog and Swiss Agency for Development and Cooperation. [7] Department of Water Resources, River Development and Ganga Rejuvenation. “Spring Census - Background.” Guidance on systematic spring and springshed mapping. [8] Swami Rama Himalayan University (26 February 2026). “SRHU Facilitates Global Workshop on Sustainable Springshed Management in Mussoorie.” [9] The Indian Express (16 September 2025). “15 killed, 16 missing in Uttarakhand after cloudburst triggers landslides, floods.” Early reporting on the Dehradun-area event and affected localities. [10] NDTV (16 September 2025). “Climate Change, Strong Winds? What’s Behind Rain Fury In Dehradun.” Report notes the IMD clarification regarding the formal cloudburst threshold. [11] The Indian Express (21-22 September 2025). “Protests in Dehradun as elevated corridor project raises more concerns following last week’s rain disaster.” Reporting on the proposed Rispana-Bindal elevated corridor and public concerns. [12] Mussoorie Dehradun Development Authority (MDDA). “Dehradun Seal/Demolition” public records of sealing, demolition and unauthorised construction actions.

[13] Mishra, K. and Garg, R. D. (2023). Study cited in reference 1; see methods and land-surface temperature results. Interpret satellite-derived surface temperature separately from air temperature. [14] Mussoorie Dehradun Development Authority. Official bye-laws page listing building bye-laws and amendments. [15] Uttarakhand Housing and Urban Development Authority. Official building bye-laws and amendments page. [16] Centre for Science and Environment. Official biography of Sunita Narain, including work on community water management and rainwater harvesting. [17] Himalayan Environmental Studies and Conservation Organization (HESCO). Official profile of Dr. Anil Prakash Joshi. [18] Tarun Bharat Sangh. Official organisation profile and information on the work of Dr. Rajendra Singh. [19] Uttarakhand Open

University. Official profile of Prof. Durgesh Pant. See also Negi and Pant (2025), reference 21. [20] G. B. Pant National Institute of Himalayan Environment. Official research profile of Dr. G. C. S. Negi. [21] Negi, G. C. S. and Pant, D. (2025). “Springs of Uttarakhand Mountains: A State of Knowledge Review with Focus on Rejuvenation of Drying Springs.” International Journal of Ecology and Environmental Sciences, 51(6), 599-620. https://doi.org/10.55863/ijees.2025.0754 [22] Central Ground Water Board. Ground Water Year Book of Uttarakhand 2023-24, issued September 2024. Consult for monitoring data and methods; verify the latest edition before quoting current levels.